



USER REQUIREMENT SPECIFICATIONS

ICOMS 2.0
Module 4 : GNC

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1.0 DOCUMENT PURPOSE

The purpose of this document is to define the User Requirements Specification (URS) for the Grid Network Control (GNC) functionalities within the ICOMS 2.0 system.

This document describes the system behaviour, workflows, and functional requirements related to the execution, control, and real-time management of outages by GNC users. It outlines the user interfaces, operational processes, and system capabilities required to support outage scheduling, authorisation, monitoring, forced outage creation, and operational handover activities.

The document serves as a reference for system development, configuration, and validation to ensure that the ICOMS 2.0 system supports the operational needs of GNC users in executing and managing outage activities within the network.

2.0 OBJECTIVES

The objective of this document is to define the functional requirements that enable GNC users to execute, monitor, and control outages within the ICOMS 2.0 system.

The key objectives include:

1. To define the system functionality required for managing scheduled, active, and forced outages by GNC users.
2. To establish the operational workflow for outage execution, including scheduling, authorisation, monitoring, and closure of outages.
3. To describe the processes for assigning, managing, and recording outage authorisations, including Taken-Active and Taken-Completed statuses.
4. To define the functionality for creating and managing forced outages with immediate approval and execution requirements.
5. To provide capabilities for real-time monitoring of outage status, including scheduled and active outage listings.
6. To support operational control activities such as outage cancellation, off-point normalisation, and status updates during execution.
7. To define the reporting and statistical capabilities required for monitoring outage execution and operational performance.
8. To provide functionality for emergency outage approval by authorised GNC users during periods when GNM users are unavailable.
9. To support shift-based operational continuity through the Handover module, including documentation, tracking, and communication of system activities.

3.0 SCOPE

This document covers the GNC user functionalities within the ICOMS 2.0 system.

The scope includes the following functional areas:

- 1. Outage Repository and Monitoring**

Viewing and managing a repository of all GNM-approved outages within the user's zone, including access to outage details, status updates, and filtering capabilities.

2. Scheduled Outage Management

Management of outages scheduled to commence, including identification of outages starting on the current day and initiation of operational actions such as authorisation or cancellation.

3. Active Outage Management

Monitoring and control of ongoing outages with active authorisations, including status updates, operational tracking, and closure of completed outages.

4. Authorisation Management

Creation, modification, and closure of outage authorisations, including recording of authorisation details, assignment of authorised personnel, and tracking of Taken-Active and Taken-Completed statuses.

5. Forced Outage Creation and Management

Creation and execution of forced outages by GNC users, including automatic approval, data entry, and integration into the operational workflow.

6. Emergency Outage Approval

Approval of emergency outages by authorised GNC users during non-operational periods of GNM, including access to dedicated approval dockets.

7. Outage Status Control and Execution

Operational control of outage lifecycle during execution, including cancellation, closure, and off-point normalisation activities.

8. Reporting and Statistics

Access to reporting and statistical functionalities shared with TOMS users, including analysis of outage execution and operational performance.

9. Calendar and Scheduling View

Access to shared calendar functionalities for visualisation of outages across timelines and operational planning.

10. Authorisation in Force Monitoring

Viewing and tracking outages with active authorisations, including status-based colour coding and filtering for operational visibility.

11. Handover Management

Management of shift-based handover processes, including documentation of operational activities, system status, outages, alarms, and instructions, with support for historical tracking and report generation.

4.0 LIST OF USERS ROLES

No	Name	Division
1		
2		
3		
4		
5		

5.0 USER REQUIREMENT

5.1 Menu Update

Legacy	ICOMS 2.0	Remark
OUTAGE STATUS		
Data Repository	Data Repository	All Approved Outages
Scheduled Outage	Scheduled Outage	All outages starting today + ongoing outages without an Authorisation
Active Outages	Active Outages	All Active outages with an Authorisation
Unresolved Outages		Removed
Forced Outages	Forced Outage	Create New Forced Outage
View/Edit Outage List	View/Edit Forced Outage	
REPORTING MODULE		
Report Page		Taken exactly from TOMS page
Statistics		
Calendar		
Off-Point Status		Removed
AUTHORISATION MODULE		
Authorisation in Force	Authorisation in Force	Shows all Authorisation that are Taken-Active
SPECIAL PAGE		
PMU Installation Management		Removed

5.2 Data Repository

This will show a listing screen of all outages within the Zone of the user account, that have an Approved status by GNM. This listing will show the status updates along with any comments by GNM, and status by GNC. The user will have access to filters to refine the listing screen to their preference.

Confirmed?	Start / End Date	Region / Station	Equipment / Description	Region / CPU / N. DC Status
☒	153283 Monday, Dec 1 2025 12:00AM Friday, Dec 5 2025 11:30PM	JOH1 / DBAY	By : Redzaan bin Samad 13233KV 90MVA T2 Additional Equipment: Job Description: Distribution Work: Testing feeder 33kv dan ROP Stability. Outage requested by DN. Pic Tuan Ibrahim. 9187752421	Agreed / Approved / Taken-Completed Highlight(s): DN Job Study Remark: TL(T1+T2)+13MVA
☒	139510 Monday, Dec 1 2025 9:00AM Tuesday, Dec 2 2025 5:00PM	JOH1 / CBRU	By : Redzaan bin Samad Busbar Coupler Additional Equipment: 132KV Main Busbar, 132KV Reserve Busbar, Job Description: Routine Maintenance, Routine Maintenance by SURD & PROA, PIC: Oas 0196555498, NCZAM 9137158186, AFIC: 0188643240, N ord# 6197734885, 1 Dec 0900 - 2 Dec 0900 BC-MBB, 2 Dec 0900-1900 BC-HBB	Agreed / Approved / Taken-Completed Highlight(s): 1) DN to be informed and to prepare CP 2) Subject PLTG normative (153396) Study Remark: TL CBRU-75MVA = 1 run BG load loss CBRU
☒	150173 Monday, Dec 1 2025 9:00AM Tuesday, Dec 2 2025 5:00PM	JOH1 / TBRU	By : SHAHRUL FITRI BIN ANNUARUSADAT 132MVACTBRU1 Additional Equipment: Job Description: Testing: TNB17212023 RMR RTU & RELAY PMU TBRU: END OF WARRANTY	Agreed / K/Ned / Pending Highlight(s): Study Remark:
☒	150166 Monday, Dec 1 2025 9:00AM Tuesday, Dec 2 2025 5:00PM	JOH1 / TBRU	By : SHAHRUL FITRI BIN ANNUARUSADAT 13233KV 90MVA T4 Additional Equipment: Job Description: Testing: TNB17212023 RMR RTU & RELAY PMU TBRU: END OF WARRANTY	Agreed / Approved / Taken-Completed Highlight(s): 1) DN to be informed and to prepare CP 2) DN to ensure rem to loading -90% Study Remark: T3+T4-50MVA
☒	153308 Wednesday, Dec 3 2025 8:30AM Monday, Dec 8 2025 5:00PM	JOH1 / KTRR	By : Wan Famp Adli Bin Wan Ahmad 275K189PMU1 Additional Equipment: Job Description: D-Sync: TRIGER SUBSTATION (NETA, SUBSTATION) END FINDER TALLY GEARBOX END TRIGER STOP TALLY STUDY D	Agreed / K/Ned / Pending Highlight(s): Before top BG outage Study Remark:

Figure 5.7a Data Repository Listing Screen

Enhancements to the Listing Screen

- Show the Restoration Time
- Show the PIC list
- Keyword will be able to search through all data
- Add filters to Outage Code (U,P,E,F); Station; Equipment Type; Job Type
- Allow users to sort by Start Date and End Date

5.3 Scheduled Outage

The active docket for GNC users, the Scheduled outage will list all the outages starting today that do not have any Authorisation. These will mark the outages GNC are scheduled to begin today.

ID	Start/End Date	Region/Station	Equipment Info	Job Description	Status	Action
153978	Monday - 29/12/2025 09:00:00 Monday - 29/12/2025 17:30:00 8 Days(s), 8 Hours(s)	KEDAH / BAHIS	Equipment: 500KV BAHIS-YOPE LINE 2 Additional Equipment:	Other(s): WAS CT and VT Termination and Testing Visits at both side. Request to open CB only at both ends. PIC- Fatima Rosli (811-5752582) Highlight: open CB only at both ends Study Remark: 1) upon outage, monitor ~5% 275KV BAHIS-YOPE ~2-5% 2) n-1 monitor, 275KV BAHIS-YOPE ~2-5% YOPE YOPN ~2-30% Distribution Work(s): Membuat Kerja Online Trip Test LUIS-RHS 33kV & Relay Maintenance T3 - T4: 0830-0900hrs PIC(s): 01825/8661 Highlight: DM job Study Remark: T3-T4-28MV	Approved	Taken Not Taken Cancel
153481	Tuesday - 30/12/2025 00:30:00 Tuesday - 30/12/2025 08:00:00 8 Days(s), 7 Hours(s)	KEDAH / KIGAR	Equipment: 13203KV ARMAA T3 Additional Equipment: 13203KV ARMAA T4	Other(s): Distribution Work(s): Membuat Kerja Online Trip Test LUIS-RHS 33kV & Relay Maintenance T3 - T4: 0830-0900hrs PIC(s): 01825/8661 Highlight: DM job Study Remark: T3-T4-28MV	Approved	Taken Not Taken Cancel
153967	Tuesday - 30/12/2025 22:00:00 Wednesday - 31/12/2025 08:00:00 8 Days(s), 8 Hours(s)	SELO / KPN3	Equipment: 13203KV ARMAA T2 Additional Equipment:	Other(s): Distribution Work(s): OFF T3, 370V REPAIR CABLE DEFECT TRACKING AT 370V BACKCABLE COMPLETION REPAIR JOINTS CABLE TO INSULATE CABLE TO TESTING & RECTIFY DEFECT PIC: MUHAMMAD HESKAM 878952685 Highlight: DM work Study Remark: T1-T2-8MV	Approved	Taken Not Taken Cancel
153229	Tuesday - 30/12/2025 22:30:00 Wednesday - 31/12/2025 11:00:00 8 Days(s), 7 Hours(s)	SELO / P107	Equipment: 13203KV ARMAA T3 Additional Equipment:	Other(s): Distribution Work(s): Solusi 38kV RHTS uk kerja kerja pembaikan pembaikan. Uluu cable end dan separatan pembaikan Engines TRHD in charge ENCR 2504FRAH +68 11 5428 3421 Highlight: DM work Study Remark: T3-33MV, T1-T2-28MV	Approved	Taken Not Taken Cancel

Figure 5.3a Listing screen of Scheduled Outages

Users will be able to Authorise or Cancel the outage from the listing screen itself. Authorising an outage will need the credentials of Authorising personnel set up by GNM in Module 1. This will be a list of users which GNC will have to pick through to authorise the outage. The user selected will receive an email with the Outage ID and details informing them of all Taken-Active and Taken-Completed status change on the selected outage.

AUTHORISATION INFORMATION			
No	Authorisation No	Time issued (24hrs:min)	Authorized Person
1	/	/	/
2	/	/	/
3	/	/	/

REQUESTER INFO		OUTAGE INFORMATION	
Requested By:	Muhammad Fatima Rosli	Date Start:	Friday, Nov 21 2025 9:00AM
Organization:	JAM JCRH	Date End:	Friday, Nov 21 2025 8:00PM
EQUIPMENT INFORMATION		Outage Code:	Unplanned
Region/Station:	KEDAH / BAHIS	Sequence:	Not Applicable
Voltage:	500	Restoration:	Immediately
Equipment:	500KV BAHIS-YOPE LINE 2	Job Type:	Others
Additional:	No Additional Equipment Requested	Description:	WAS CT and VT Termination and Testing Visits at both side. Request to open CB only at both ends. PIC- Fatima Rosli (811-5752582)
Daily PTW:	As Requested		
CT Rate:	No CT Ratio data or setting provided		
STUDY REMARK / STATUS:			
OPN's Status:		Approved / (modul substation line break)	
Highlight(s):		open CB only at both ends	
Remark:		1) upon outage, monitor ~5% 275KV BAHIS-YOPE ~2-5% 2) n-1 monitor, 275KV BAHIS-YOPE ~2-5% YOPE YOPN ~2-30%.	
NLEDC Status:		Taken Active	
Remark:			
Date:		Tuesday, December 30, 2025	

Figure 5.3b Adding Authoriser

In order to add or close and authorisation, the GNC user must enter the Authorisation No (taken from the from filled out on site), the date and time, select an Authoriser from the dropdown list and add a remark.

This will create a clean record of all **Taken-Active** and **Taken-Completed** status for each outage. Upon authorisation, the system will automatically assign a **Taken-Active** status to the outage. Upon Authorisation being closed, the system will shift the status to **Taken-Completed**. This process can be repeated multiple times before the outage is marked **Outage Closed**.

5.4 Active Outages

Once an outage has been given an Authorisation status, they will shift to the Active Outages listing screen. This listing will show all currently open outages with an Authorisation. 'This can be either **Taken-Active** or **Taken-Completed**

ID	Scheduled Date	Region/Station	Equipment Info	Job Description / Study Remarks	Action
11049	Sunday - 27/01/2022 08:00:00 Wednesday - 26/01/2022 17:00:00 1375 Days(s), 9 Hours(s)	TSBG / TBLG	275W000MT KL01 Additional: 275W Reserve Number: 275W000MT KL02	Testing(s) To dismantle primary equipment from 275W system for both TBLG L15 & L25. SUREC will proceed for other related work after outstanding completed. RSI will be taken (27 Mar - 0900-1400) and with both line at the same time. Highlight: 1) PWSM & ESTL to be informed. 2) Spar. Study Remarks: Can arrangement KNYS-0MW, HTSG-0MW, TL PWSM-Abs load, ESTL-0MW if rem RD Rtp. PAKA-KAWA-242 0% PAKA-SGT-2050%, will drop @ R00K-6.85p.	Complete
147270	Tuesday - 26/04/2022 18:00:00 Wednesday - 26/04/2022 17:00:00 240 Days(s), 7 Hours(s)	SELG / OLPT	275W000MT KL01 Additional: 275W000MT KL02	Project(s) The outage is needed for New Substation PMU BTIC project and connection from BTIC to OLPT. The scope of the outage is as follows: 1) Foundation works for new towers under the 275W OLPT/ESTL/Rp. Rwp To decommission the line after the commissioning of 275W OLPT-BTIC. Thank you. PIC: Maude - 0163856443 Chua - 011-18313814 Highlight: Line on work. Study Remarks: None.	Complete
151386	Monday - 26/09/2022 19:30:00 Thursday - 19/10/2022 17:00:00 450 Days(s), 7 Hours(s)	PRWG / BTCH	275W150V 150MVA SGT3 Additional:	Project(s) Outage SGT3 for dismantling cable-bell and CSE. Based on discussion between OD, OM, OSO and GSE on 16 Apr 2024, the SGT3 will be in outage until new 150V RSI is ready to be commissioned. PR: Hant (0197520346). N-gate (019 54 422 3300). Highlight: SGT3 on work and will be long outage for 1 year (based on OM&G's info). Study Remarks: None.	Complete
158870	Friday - 03/04/2022 09:00:00 Sunday - 04/04/2022 17:00:00 93 Days(s), 17 Hours(s)	KEDP / PHLH	13233 AGMVA T1 Additional:	Project(s) Project(s): TMR2460001: TTC 1) Replacement & Refurbishment for 132W Transformer No.1 Box 2) Increase Transformer Capacity from 0MVA 13233W T1 to 50MVA 13233W T1 PIC: a) Hantah (02-7149) 015-2618559 Highlight: 1) OM informed and prepare cost plan (CP). 2) OM confirmed T2 and T3 health index are good and no defect. Study Remarks: T1-13233WVA-40MVA T2-50MVA-10MVA	Complete
153545	Monday - 01/12/2022 09:00:00 Wednesday - 26/12/2022 18:00:00 10 Months, 16 Hours(s)	PRWG / DMTS	50MVA150V T2 Additional:	Project & Refurbishment(s) Outage is required for work related to RCT2 (rehab & refurbishment works at ATWV 2 - CB 22130 & 22230 are req and for the substation 100% Fund 019 515 8845. 0 from 019 515 8845. 0 from 019 515 8845. 0 from 019 515 8845.	Complete

Figure 5.4a Active outage listing screen

Users will be able to set outages to Outage Closed status by the use of the one click button. The system will prompt the user to normalise the off-points. If "Yes" is selected, all off-points effected by the outage with a closed status will change to open and vice versa.

If an Off-Point is being affected by multiple outages, the system will normalise all other off-points and request a reconfirmation for the one/multiple off points used by other outages. Each off-point effected by other outages will need to be confirmed one by one.

5.5 Forced Outage

The GNC users are the only roles that are allowed to create a Forced Outage. Each created outage is automatically given an **Approved** status and parked in the Scheduled Outage listing for the operation to commence.

Field	Field Type	Remarks
Zone	Dropdown	Zones Created by Admin
Organisation	Dropdown	All Organisations in the system
Station	Dropdown	Stations within the selected Zone
Voltage	Dropdown	Voltage across the system
Equipment Type	Dropdown	Equipment types for the Voltage Selected
Line Filter	Dropdown	Single, Tee-Off, Quad
Equipment	Dropdown	All Equipment of the Voltage and Type selected within the Station
Additional Equipment Type	Dropdown	Equipment types for the Voltage Selected. Also have the All option to display all Equipment within the station
Additional Equipment	Multi-Select	All Equipment for the additional type selected. Users can multi-select multiple equipment across types

Date Start and Time	Calander, Dropdown	Default time set to 0800, dropdown will show increments of 30 minutes
Date End and Time	Calander, Dropdown	Default time set to 1700, dropdown will show increments of 30 minutes
Code (Outage parameters)	Text Field	Auto filled with F, denoting forced outage
Job Type	Dropdown	Job Types created by Admin
Approved By	Dropdown	List of GNC seniors ranks.
GNC Status	Dropdown	Will automatically be set to Approved upon creation, pending the Approved by Field is filled
Description	Text Box	No character limit, open text box for users to fill as needed.
PIC	Dropdown	List of Requestors in the system
Requestor Notes	Text Field	PIC Requestor selected can add outage details once outage is created.

Table 5.5a Fields for creating new Forced Outage

After outage creation, the PIC (REQUESTOR) selected during the outage creation process can view the outage and add additional notes or information relating to the outage.

5.7 Outage Approval

During the weekends and long holidays, GNM users will not be available to Approved emergency outages that are created by REQUESTORS. In this case, a sub role of the GNC users with the right permissions set up by the GNM ADMIN will be responsible for outage Approval. They will have access to an Approval Docket similar to the one in Module 3 – TOMS; however, this listing will show only Emergency Outages.

The GNC user will then be responsible for setting the status to Approved before the outage can be carried out. There will be an additional Approved By hardcoded dropdown with a list of approval personnel positions which will need to be selected before approval can commence.

5.8 Statistics

GNM and GNC users will share the same statistics and reports. Please refer to Module 3 – TOMS for further details

5.7 Calander

GNM and GNC users will share the same Calander. Please refer to Module 3 – TOMS for further details

5.8 Authorisation in Force

Once Approved by GNM, outages can be worked on by the GNC team. This is done by Authorising the outage to give it a Taken-Active status. If the Authorisation is closed, the status changes to Taken-Complete. GNC can add and remove Authorisations on outages at will, as they work over the period of the outages.

The Authorisation in Force listing will show all the Outages that currently have a Taken-Active status.

The listing screen will be colour coded to show the timing of the outage. Green will be those are currently in the outage timeline window. Yellow will be for those that have been extended by GNC users. Red will denote the outages where the original time has been over, but they still have a Taken status, and have not been closed with any of the Outage Closed status.

Region		AS	Keywork	Month	Custom Month	Year	Select Year	Start	Start Date	ASC	Duration	Check	Excel
Total No of Outage In-Force found based on search criteria : 179 / Expired authorisation count as of 11/06/2018: 148													
ID	Date Info	Duration (Days)	Region / Station	Equipment Info	Job Description	Action							
0019	Monday, Jan 8 2018 12:00PM Deadline: Jan 31 2018 12:00AM Code : E	900.5	TELUGU / TELUGU	27100000000000 Additional Equipment 27100000000000	Station : Taken-Active Disconnection of supply under SCADA (Operate Wayah 2) instruction	Review							
01416	Tuesday, Jan 12 2018 10:00PM Deadline: Dec 31 2018 12:00AM Code : E	1003.1	KULUM / SHEN	1230V GCB G12 200 Additional Equipment N/A	Station : Taken-Active SLA Expired. To disconnect generator	Review							
01416	Tuesday, Jan 12 2018 10:00PM Deadline: Dec 31 2018 12:00AM Code : E	1003.1	KULUM / SHEN	1230V GCB G12 200 Additional Equipment N/A	Station : Taken-Active SLA Expired. To disconnect generator	Review							
01417	Wednesday, Apr 21 2018 10:00PM Deadline: Dec 31 2018 12:00AM Code : E	505.1	KULUM / SHEN	27100V GCB G13 M20 Additional Equipment N/A	Station : Taken-Active SLA Expired. To disconnect generator	Review							
73364	Thursday, Mar 8 2017 17:00AM Deadline: Feb 28 2018 17:00AM Code : F	356	KULUM / KULUM	1230V GCB G12 200 Additional Equipment N/A	Station : Taken-Active SLA Expired	Review							
73364	Thursday, Mar 8 2017 17:00AM Deadline: Feb 28 2018 17:00AM Code : F	356	KULUM / KULUM	1230V GCB G12 200 Additional Equipment N/A	Station : Taken-Active SLA Expired	Review							
73364	Thursday, Mar 8 2017 17:00AM Deadline: Feb 28 2018 17:00AM Code : F	356	KULUM / KULUM	1230V GCB G12 200 Additional Equipment N/A	Station : Taken-Active SLA Expired	Review							
01495	Thursday, Aug 24 2017 3:00PM Deadline: Dec 31 2018 3:00PM Code : E	494.3	PRONG / GMB	1230V GCB G11 100 Additional Equipment 1230V GCB G11 100	Station : Taken-Active Supply disconnection	Review							
04955	Sunday, Apr 15 2018 12:00AM Deadline: Dec 31 2018 12:00AM Code : E	200	KULUM / SHEN	1230V GCB G11 100 Additional Equipment N/A	Station : Taken-Active SLA Expired. To disconnect generator	Review							
04955	Sunday, Apr 15 2018 12:00AM Deadline: Dec 31 2018 12:00AM Code : E	200	KULUM / SHEN	1230V GCB G11 100 Additional Equipment N/A	Station : Taken-Active SLA Expired. To disconnect generator	Review							
0000	Sunday, Apr 15 2018 12:00AM	0	1230V GCB G11 100	Additional Equipment N/A	Station : Taken-Active	Review							

Figure 5.12a Listing screen of Authorisation in Force

Each Authorisation listing screen will have filters for Region, Month, Year, Start Date, On Time/Extended/Overdue and the ability to sort by Ascending or Descending based on start date.

5.10 Handover

The Handover module is a new addition to the ICOMS 2.0 system, creating a live page showing the latest information of activities across the system for a new shift to peruse though, and a leaving shift to leave remarks and information.

There are three shifts per day for GNC users, Morning, Evening, and Night. For ease of access and readability, there will be a multi-layered tab system showing each parent tab of Morning, Evening and Night with the multiple tabs as specified below detailing the Handover information

Tab	Field	Field Type	Remark
Handover Control	Control manager	Dropdown	Dropdown Of Users
	Switch Engineer	Dropdown	Dropdown Of Users
	Switching engineer 2	Dropdown	Dropdown Of Users
	Despatcher	Dropdown	Dropdown Of Users
	Control Assistant	Dropdown	Dropdown Of Users
	Date	Auto Filled	
Outages	kV		Shows different outages in the different kV levels
	Date		
	Outage Equipment		Equipment of the Outage
	Tripping		All forced outages with the job type Tripping will be shown
	Changes in Off Points/Violations and CA		Auto display of Off-Points + Manual Entry by user
	Remarks		Manual Entry
Check Alarm List and Click Report	Date		
	PMU		
	Alarm	Manual Entry	
	PIC	Manual Entry	

Emergency/Unplanned Outage			
System Monitoring	Fault Level Violations		
	Oil units		
	Distillate Units		
	PGL T/Line Export		
	LDS Status		
	LSS Voltage Control		
Commissioning			
Outage Complete	Off Point Normalisation		
Special Protection Scheme	Station Name		
	Detail	Manual Entry	
Special operation	Date		
	Equipment and Job Type		
	Detail	Manual Entry	
Other instruction	Date		
	Title		
	Detail		

Table 5.10a Tabs and Fields for Handover

The Handover details will be filled out by the selected Roles set by the GNM ADMIN. These fields will be shown as an editable table. Users will have the functionality of manually adding new fields if necessary.

Once the fields are set, the user can then “Pass Handover” to the next shift. A copy of the data will be transferred to the next shift screen, and the current shift will no longer be editable. The Handover document can be exported out as an Excel or PDF.

Users can create a customized Handover report, where they have the ability to remove outages before printing to PDF/Excel.

The system will store old Handover details, allowing the users to search past Handover notes by selecting the Date and Shift fields.

5.10.1 Sample Screens

Export

Pass Handover

Daily Shift Log

Monitoring and operational handover for control center operations.

Morning

Evening

Night

Handover Control

Outages

Emergency Outages

System Monitoring

Commissioning

Protection Scheme

Special Ops

Other Instructions

Handover Control Details

LIVE MODE

LAST SYNCED: 2M AGO

FIELD	VALUE (EDITABLE)	REMARK
Control manager	Alexander Vance	Add optional remark...
Switch Engineer	Sarah Jenkins	Acting as primary supervisor for Zone A
Switching engineer 2	Michael Thorne	—
Dispatcher	Elena Rodriguez	Remote coordination active
Control Assistant	James P. Sutherland	—
Date	2023-10-24	Automatic timestamp sync active

Figure 5.10.1a Example of Handover Screen

Outage Logs

Review and update current equipment downtime and trip status.

Morning

Evening

Night

Outages

Equipment State

Relay Signals

Substation Loads

Active System Outages

3 TOTAL OUTAGES

KV	DATE	OUTAGE EQUIPMENT	TRIPPING	REMARK	ACTIONS
220kV	2023-11-24	T-04 Main Transformer		Differential protection acted during ligh...	
110kV	2023-11-24	Bay L-12 Bus Coupler		Scheduled insulator cleaning and verifi...	
33kV	2023-11-23	Feeder F-09 Substation B		Persistent earth fault detected on phas...	

Append New Log Entry

Figure 5.10.1b Example of Handover Screen

Special Operations

High-priority field maneuvers and sensitive equipment adjustments scheduled for the current shift. Ensure all safety protocols are logged.

[+ Add Special Operation](#)

Shift Log: Special Ops

REAL-TIME FEED



DATE	EQUIPMENT	JOB TYPE	DETAILS	REMARK
Oct 24, 2023 - 08:30	T-900 Turbine Unit	CALIBRATION	Sensitive recalibration of the core vibration se...	Monitor for 2 hours
Oct 24, 2023 - 09:15	Switchgear Panel A4	MAINTENANCE	Emergency bypass testing for the secondary l...	Clearance obtained
Oct 24, 2023 - 10:45	Valve Controller V2	INSPECTION	Manual override check for special flow require...	Ongoing...

[+ Click here to add a new operational entry](#)

Showing 3 of 12 Special Operations

[View Archived Ops](#) [Export Logs](#)

Figure 5.10.1c Example of Handover Screen

Handover Management
Live overview of shift activities and transfer notes

10/04/2026

Excel

PDF

Pass Handover

SHIFT SELECT

Morning

Evening

Night

Handover Control

Outages

Tripping

Emergency/Unplanned Outage

System Monitoring

Outage Complete

Special Protection Scheme

Special operation

#

CONTROL MANAGER

SWITCH ENGINEER

SWITCHING ENGINEER 2

DESPATCHER

CONTROL ASSISTANT

DATE

No records found for this section.
[+ Add the first entry](#)

+ Add Row

Figure 5.10.1d Example of Handover Screen

5.10.2 Exported Report

NLDC Control Manager Handover Notes

	Control Manager	Switching Engineer I	Switching Engineer II	Despatcher	Control Assistant
Name:	Hafiz	Syarmizal	Asymam	Syabil	Edy
Date :					
Shift :	Evening				

	Date	Outages	Tripping	Changes in off points / Violations & CA	Remarks
500kV			Forced outage with job type Tripping -	Auto offpoint - Manual	Manual
			Pull all regardless of status – kept for 1 day after status is Outage Completed		
275kV					
		Only taken-active and taken - completed			
132kV					
Cap Bank, Reactor, SVC	--Hard coded equipment				
Generation					

CHECK ALARM LIST & CLICK REPORTED (OWN & 1 PREVIOUS SHIFT)					
Critical Alarm Reporting	SF6, Bucholz, PRD, Oil Lvel, OTI, WTI etc. (Don't remove until alarm cleared at NLDC alarm list)				
Date	PMU	Alarm		PIC	
		All manual			
Emergency / Unplanned outage :					
Station + equipment	Manual				
System Monitoring					
Fault level Violations	Manual				
Oil Units					
Distillate Units					
EGAT HVDC					
PGL T/Line Export					
LDS status					
LSS Voltage Control					
Commissioning :					
Station + equipment + kV					
Outage Completed :					
Outages closed last 1 days	Off-Point Normalization :				
	Yes/No + Remarks				
	--This list is only for outages with off points				
SPS (Special Protection Scheme) :					
Manual	Have x and + -- conformation required on x				

User #5

Signature:

Name:

Designation:

Dept.:

Date:

User #6

Signature:

Name:

Designation:

Dept.:

Date: